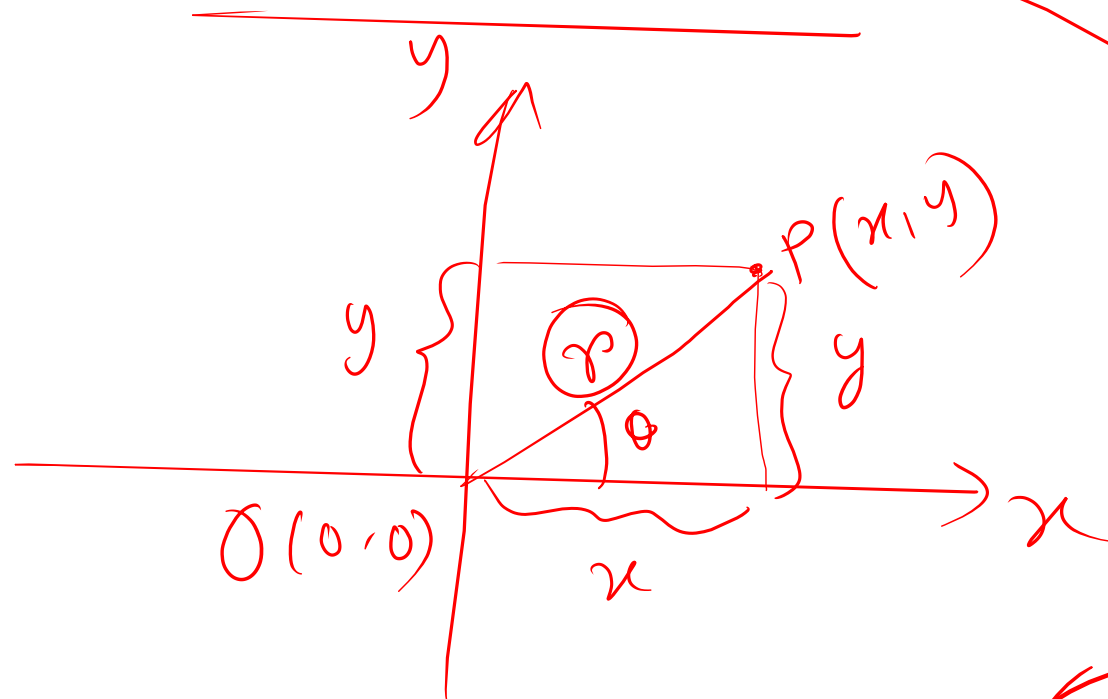


2D

Cartesian



$$x = r \cos \theta$$

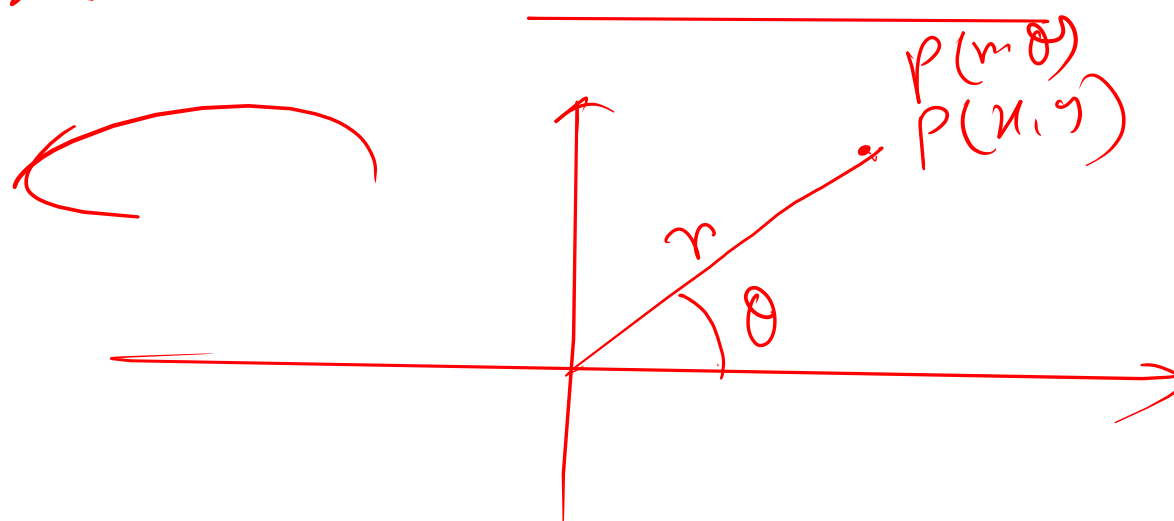
$$y = r \sin \theta$$

$$r = \sqrt{x^2 + y^2}$$

$$\tan \theta = \frac{y}{x}$$

$$\Rightarrow \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

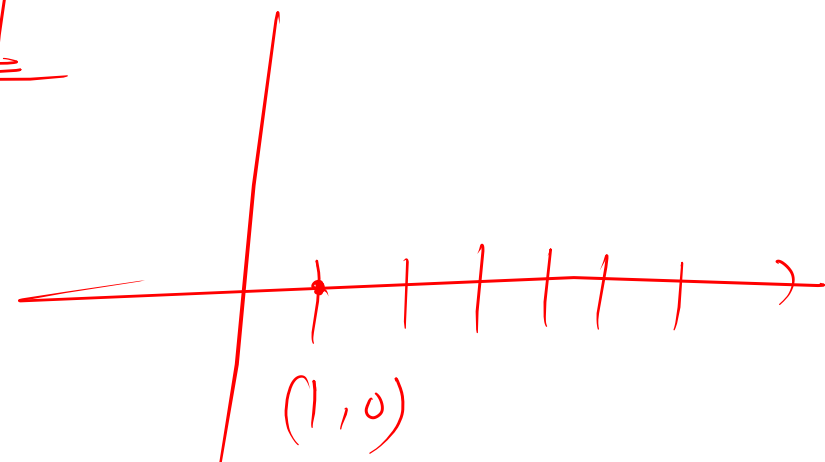
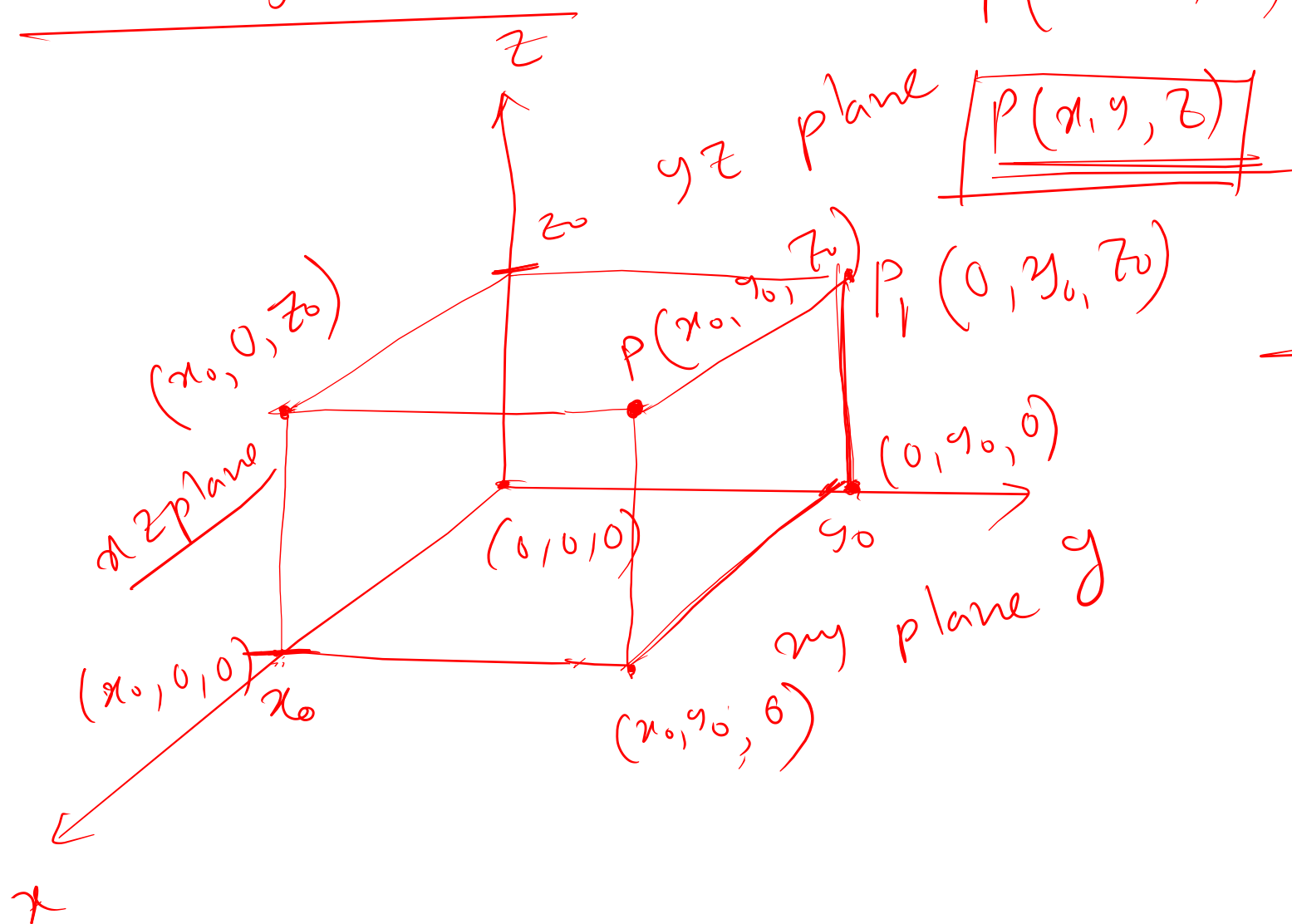
Polar



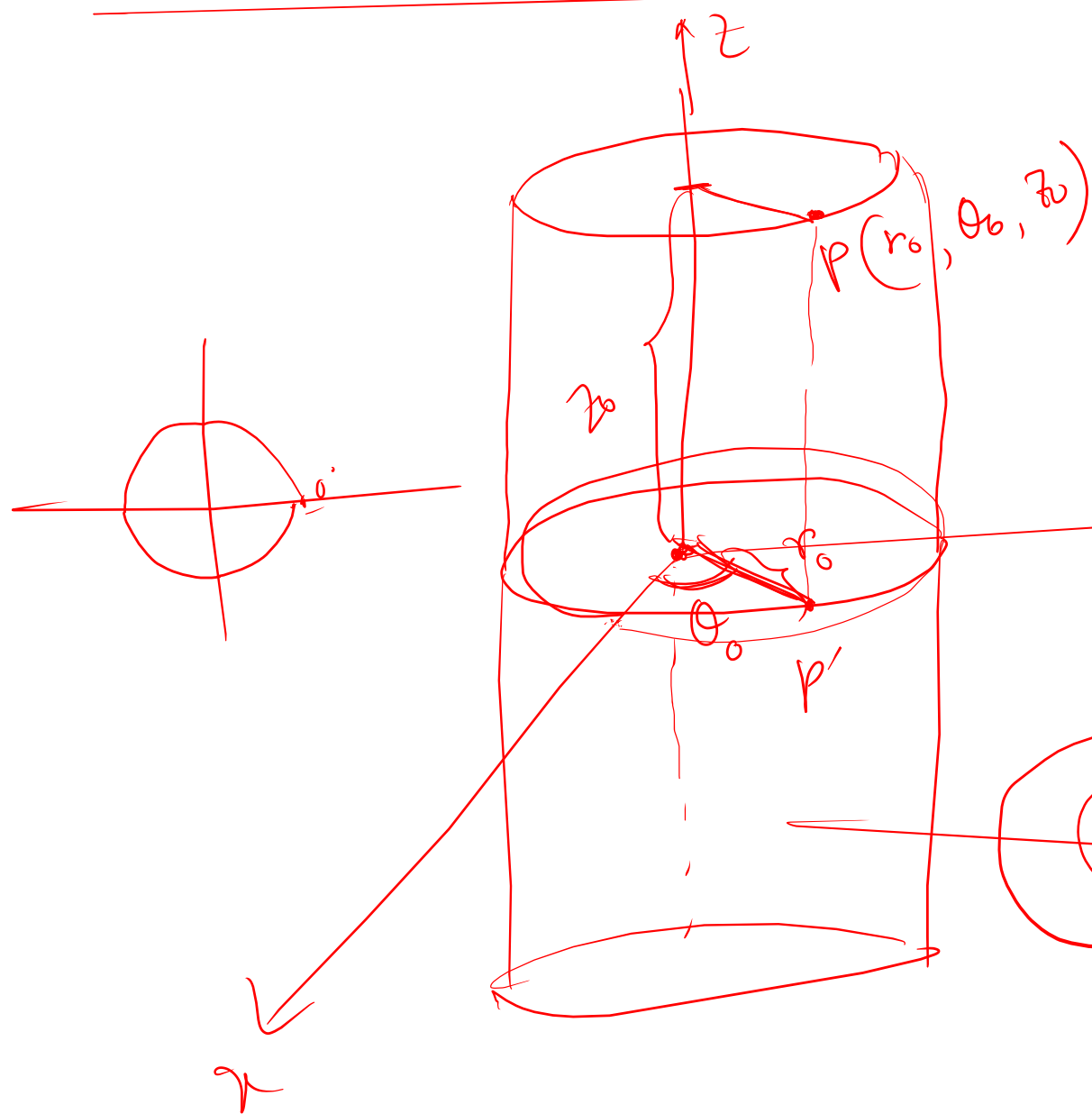
3D
Rectangular

Rectangular

$$P(x_0, y_0, z_0) \rightarrow \overline{\overline{x, y, z}}_{(-\infty, \infty)}$$



Cylindrical Coordinate System:



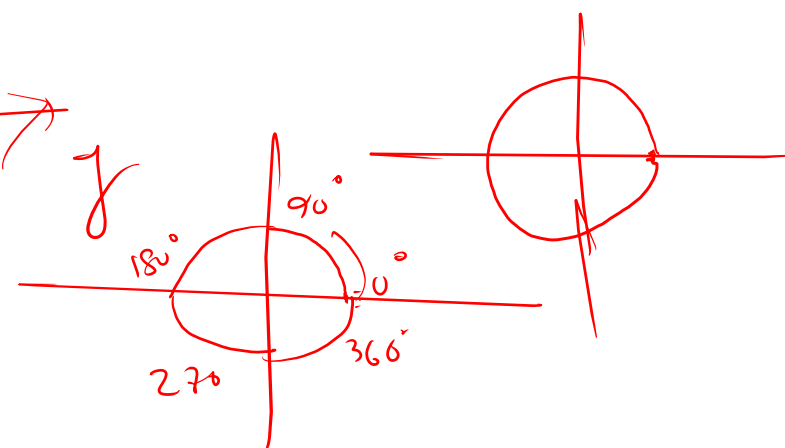
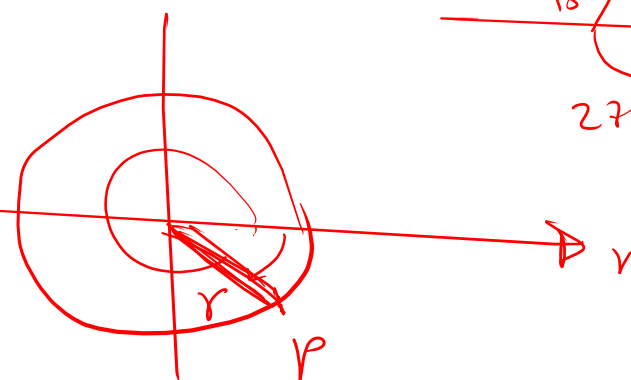
r_0, θ_0, z_0

$$r = \sqrt{x^2 + y^2}$$

$$\boxed{r \geq 0}$$

$$\boxed{P(r, \theta, z)}$$

$$0 \leq \theta < 2\pi$$



Spherical Coordinate System!

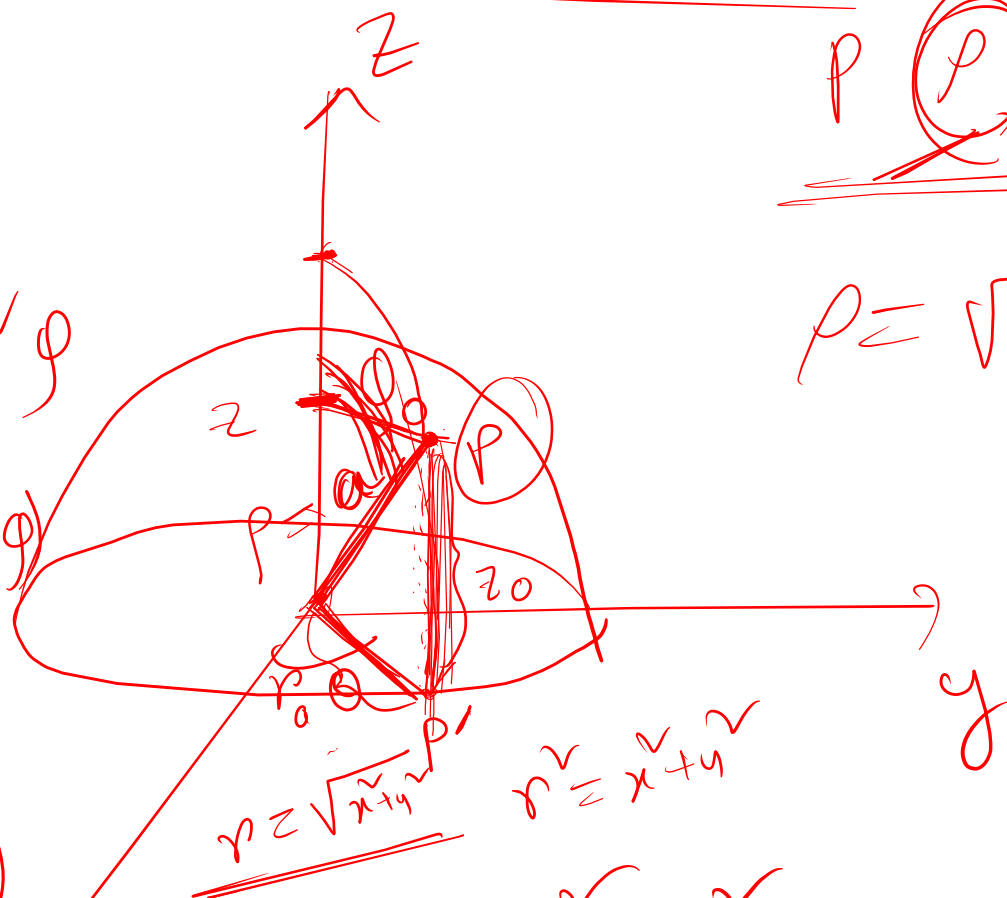
$$\begin{aligned}\vec{\rho} &= \vec{r} + \vec{z} \\ \Rightarrow \nabla \vec{r} &= \vec{\rho} - \vec{z} \\ \Rightarrow \nabla \vec{r} &= \vec{\rho} - \vec{\rho} \cos \varphi\end{aligned}$$

$$\Rightarrow \nabla \vec{r} = \vec{\rho} (1 - \cos \varphi)$$

$$\Rightarrow \nabla \vec{r} = \vec{\rho} \sin \varphi$$

$$\Rightarrow \nabla r = \rho \sin \varphi$$

r



$$\begin{aligned}r &= \sqrt{x^2 + y^2} \\ r^2 &= x^2 + y^2\end{aligned}$$

$$\vec{r} + \vec{z} = \vec{\rho} = \vec{r}$$

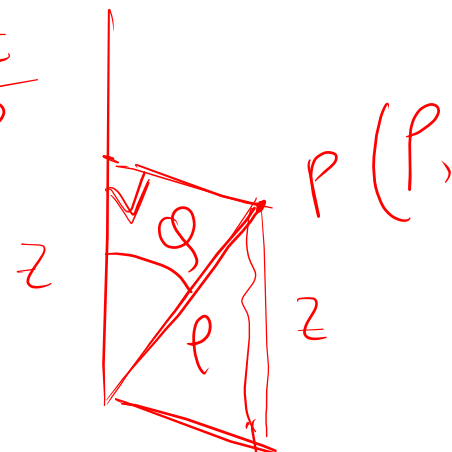
$$\Rightarrow \rho = \sqrt{r^2 + z^2} = \sqrt{x^2 + y^2 + z^2}$$

$$\Rightarrow \rho = \sqrt{x^2 + y^2 + z^2}$$



$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$$\cos \varphi = \frac{z}{\rho}$$



$$r = \rho \sin \varphi$$

$$z = \rho \cos \varphi$$

z

$$\begin{aligned}z &= \rho \cos \varphi \\ \rho &\geq 0\end{aligned}$$

$$0 \leq \varphi < 2\pi$$

$$\cos \theta = \frac{\text{Base}}{\text{Hypotenuse}}$$

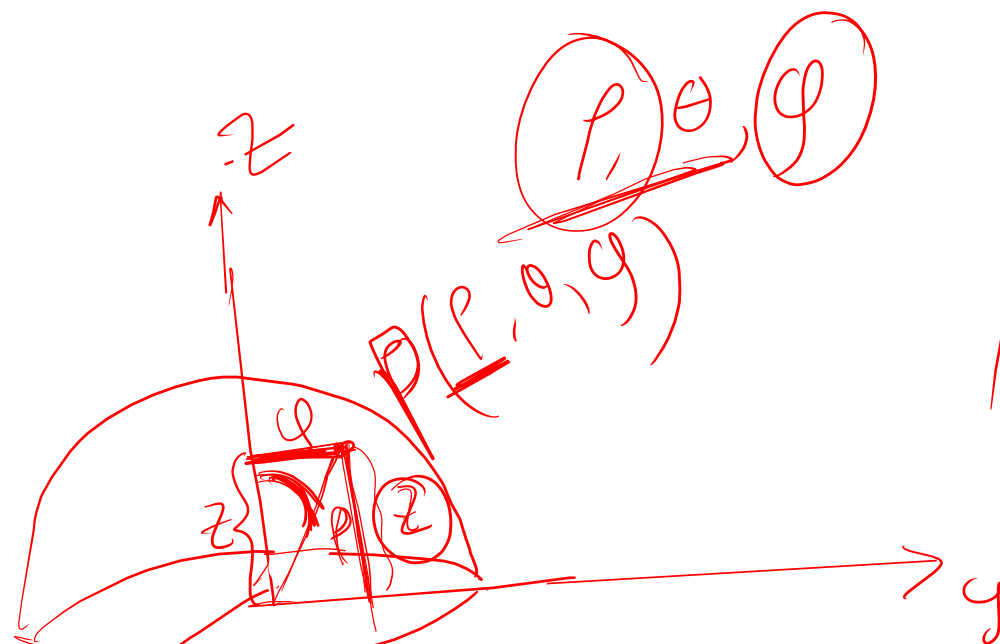
$$\cos \phi = \frac{z}{\rho}$$

$$z = \rho \cos \phi$$

$$x = \rho \sin \theta$$

$$y = \rho \sin \theta$$

ρ



$$\rho \geq 0$$

$$0 \leq \theta < 2\pi$$

$$0 \leq \phi < \pi$$

Rectangular to Spherical

$$\rho = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \tan^{-1} \frac{y}{x}$$

Spherical to Rectangular

$$\phi = \cos^{-1} \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

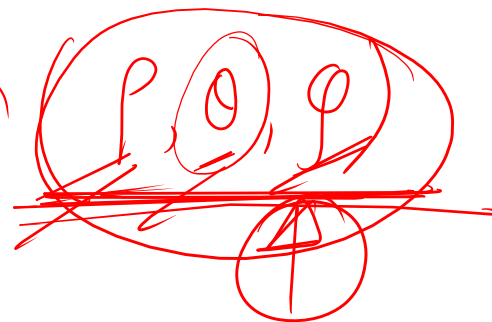
$$x = \rho \sin \theta \cos \phi$$

$$y = \rho \sin \theta \sin \phi$$

$$z = \rho \cos \phi$$

Cylindrical to Spherical

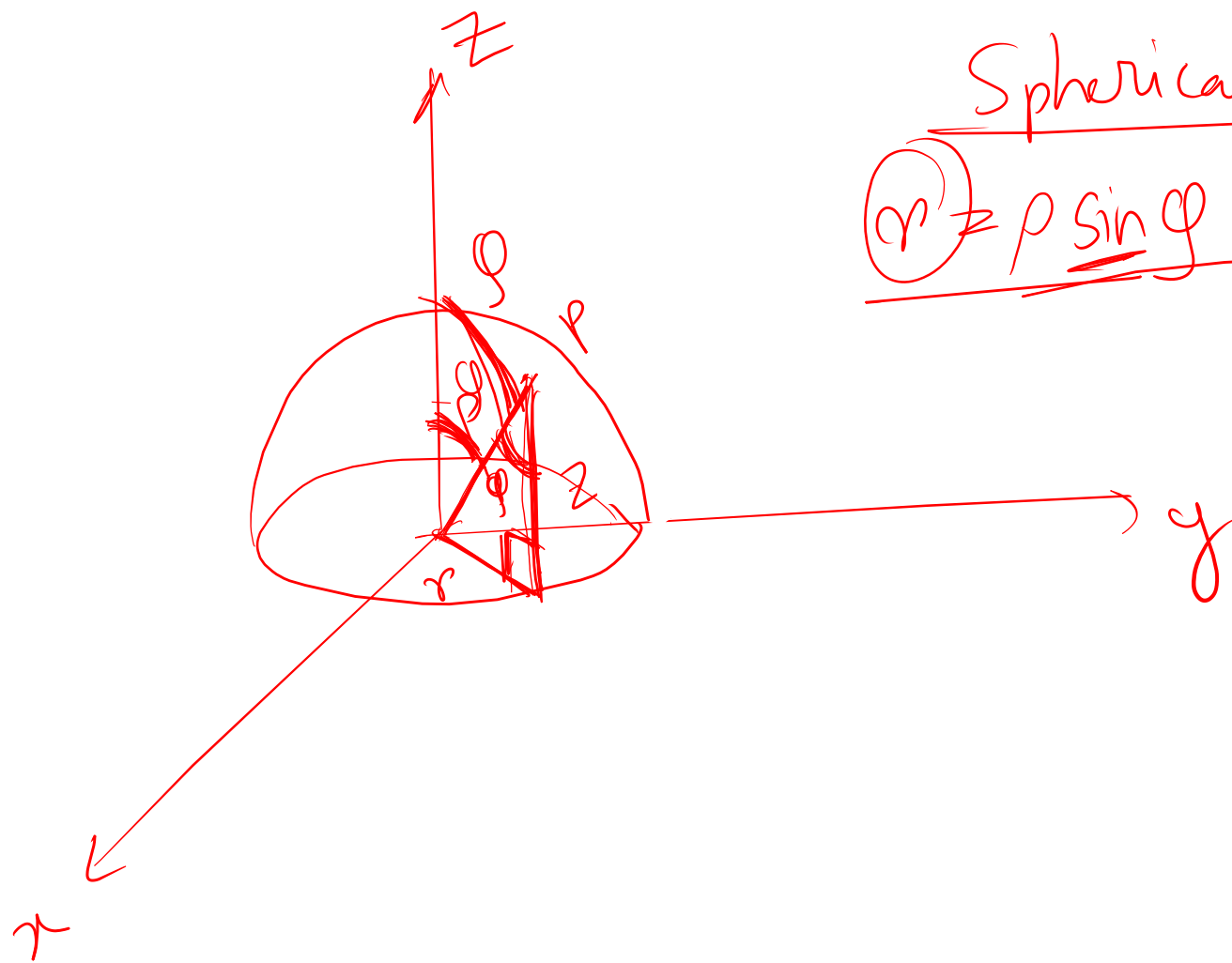
~~(r, θ, z)~~



$$\underline{\underline{\rho^2 = r^2 + z^2}}$$

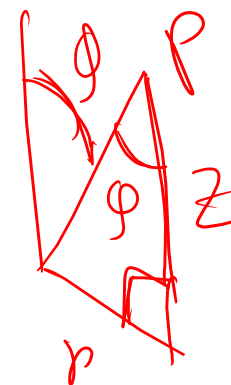
$$\underline{\underline{\theta = \theta}}$$

$$\tan \phi = \frac{r}{z}$$



Spherical to Cylindrical

$$\underline{\underline{r = \rho \sin \phi}}, \quad \underline{\underline{\theta = \theta}}, \quad \underline{\underline{z = \rho \cos \phi}}$$



Convert Rectangular to Spherical & Cylindrical.

$$\begin{pmatrix} 1, -\sqrt{3}, -2 \end{pmatrix}$$

$x \quad y \quad z$

Spherical:

$$(p, \theta, \varphi) \quad p = \sqrt{x^2 + y^2 + z^2} = \sqrt{1 + 3 + 4} = \sqrt{8}$$

$$\theta = \tan^{-1}\left(\frac{-\sqrt{3}}{1}\right) = -\frac{\pi}{3}$$

$$\varphi = \cos^{-1} \frac{-2}{\sqrt{1+3+4}} = \cos^{-1}\left(\frac{2}{\sqrt{8}}\right) = \cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) = \frac{3\pi}{4}$$

Cylindrical (r, \theta, z)

$$r = 2, \quad \theta = -\frac{\pi}{3}, \quad z = -2$$